

Cell-State Reachability

Built with Claude

Life-Sciences Hackathon · Research / Lab track

Can knockdown point a cell where you want it to go?

THE REFRAME

Most tools rank what a perturbation might do. This asks an earlier question — can non-negative combinations of a screen's measured effects point the transcriptome toward the target — and it certifies when they provably cannot.

FLAGSHIP · Th2 → Th1, resting primary human CD4⁺ T cells

0.448

held-out cosine

in-sample 0.627 · the honest,
held-out-gene score

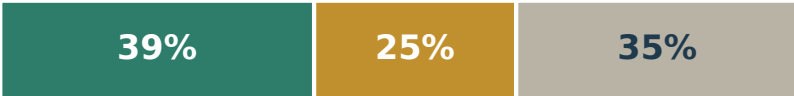
VERDICT: PARTIALLY REACHABLE

> all 60 shuffled targets
empirical $p = 1/61$ · $z \approx 24$

KKT optimality violation
 1.1×10^{-11} (cone optimality)

SIGNED DECOMPOSITION

of the target direction



LOF 39% — reachable by knockdown

GOF proxy 25% — sign-flipped effect

neither 35% — unmatched by either proxy

knockdown is never the majority modality (atlas mean LOF 0.34)

FOUR OUTPUTS, EVERY RUN

① Directional verdict

held-out + null-calibrated,
not a similarity ranking

② Ranked sparse panel

greedy mixture → compact
knockdown candidate set

③ Dual certificate

Farkas/KKT proof when the
target is outside the cone

④ Unmet-readout list

ranked CRISPRa / de-repression
hypotheses — to be tested

SAME OPERATOR, NO RETUNING

Transfers unchanged to a Norman K562 CRISPRa screen (held-out CEBPA state, cosine 0.878). Additivity guardrail: 126 doubles, median cos 0.71 (bounded, not exact).

SCOPE

Directional feasibility relative to one measured screen — not functional rescue. Combinations assume bounded additivity.

Recovered regulators (GATA3↓, TBX21↑) are positive controls, not discoveries. Every nomination is a wet-lab hypothesis, never a validated target.